Assignment 1: Portfolio | Assignment Type: Group Project | Weightage: 20%

Weight	20% of final grade
Group Size	4 students per group (as per LMS)
Ethics Declaration	Required at Group Level. Every group must submit a signed Ethics and Integrity Declaration Form before work begins. See Section 1.6.
Preferred Submission Format	Jupyter Notebook (.ipynb), GitHub Repository, and Trained Model File: 7% Project Report (PDF, 2500 words): 8% Demo Video (5–7 min) or Live Demo: 3% Dataset Source Documentation: 2%
Due Date	8th October 2026

1.1 Course Learning Outcomes Mapped

This assignment addresses the following course learning outcomes (CLOs).

CLO A: Explain fundamental concepts, scope, and applications of data science and AI.

CLO C: Analyze data using statistical and visualization methods to derive meaningful insights.

1.2 Assignment Overview

This is a team effort. Your group will choose one of the problem statements listed below, find and work with a real-world dataset, explore it thoroughly, build a predictive or analytical model, and present your findings in a structured report. You are also expected to reflect on the business or social value of your work.

The goal is to go through the full data science lifecycle as a team. That means dividing responsibilities thoughtfully, reviewing each other's work, and arriving at a coherent final output together.

1.3 Choose One Problem Statement

Your group must choose exactly one of the following four problem statements. Each comes with a suggested dataset link, but you are free to use a different version or an enhanced dataset if you can justify why it is more appropriate.

Option C: Identifying Patterns in Global E-Commerce Sales

Online retail generates data at a scale that would have been unimaginable twenty years ago. Every click, every purchase, every return creates a record. Most of that data sits in databases and is never looked at carefully. When someone does look at it carefully, they often find patterns that change how a business operates.

Your group will work with a large transaction dataset from a real UK-based online retailer. It covers millions of rows across multiple years and includes product descriptions, quantities, prices, customer identifiers, and country codes. The first thing you will notice is that this is not a clean, ready-to-use dataset. There are cancellations mixed in with purchases, negative quantities, missing customer IDs, and product codes that do not follow any obvious logic. Getting this data into a form you can analyse is itself a meaningful piece of work.

Once you have clean data, think about the different questions a business might actually care about. Which products are the most valuable and why? Is it because they sell in high volume, or because they have a high value per transaction, or because they are purchased repeatedly by loyal customers? Are there seasonal patterns in what people buy? Which countries generate the most revenue, and is that just because they have more customers, or do customers in certain regions spend more per order?

The main modelling task is customer segmentation using K-Means clustering. Construct meaningful customer-level variables that capture differences in purchasing behaviour. For example, you may use RFM variables — Recency, Frequency, and Monetary Value — as inputs to the clustering model.

After creating the clusters, analyse what distinguishes each customer segment and what those differences mean for the business. A segment is only useful if it leads to a different action. Saying that high-value customers spend more is not an insight. Saying that a particular segment purchases frequently but has recently become inactive, and may therefore benefit from a re-engagement campaign, is more actionable.

Seasonal sales patterns may be explored as part of your exploratory analysis, but time-series forecasting is not required for this project.

Your group should also think about what this data cannot tell you. You have transactions, but you do not have browsing data, marketing spend, or return rates. How does that limit what you can conclude?

Dataset: https://archive.ics.uci.edu/dataset/502/online+retail+ii

Mandatory technical requirements: Data cleaning at scale including handling cancellations and missing values, construction of meaningful customer-level features such as RFM variables, customer segmentation using K-Means clustering, interpretation and profiling of

the resulting clusters, at least five well-labelled visualisations that tell a coherent story, a business insights section with actionable recommendations, and a discussion of what the data cannot tell you.

1.4 Deliverables

(a) **Jupyter Notebook (.ipynb):** Clean, well-commented code covering the full workflow from loading data to model evaluation. Outputs should be visible without needing to re-run the notebook.

(b) **Trained Model File:** Export your best model as a .pkl or .joblib file using scikit-learn's joblib or pickle library.

(c) **Project Report (PDF, 2500 words):** A structured written document. See Section 1.7 for what to include.

(d) **GitHub Repository:** A public or shared repository with all code, the notebook, and a clear README.md explaining how to run your project.

(e) **Dataset Source Documentation:** A short document or README section listing where your data came from, when you accessed it, and any preprocessing steps you applied before analysis.

(f) **Demo Video (5 to 7 minutes) or Live Demo:** Walk through your notebook, explain your findings, and show your model working on a test case.

1.5 Project Report Structure

Your 2500 word report should follow this structure. Word counts are approximate guidelines, not hard limits.

(a) **Cover Page:** Project title, group members and IDs, chosen problem statement, date of submission.

(b) **Executive Summary:** Summarise what you did, what you found, and what the key takeaway is.

(c) **Introduction:** Describe the problem, why it matters, and the dataset you used.

(a) **Data Description and Preprocessing:** What does the dataset look like? What issues did you find and how did you fix them?

(b) **Exploratory Data Analysis:** What patterns, trends, or anomalies did you discover? Reference your visualisations.

(c) **Modelling Approach:** What models did you use and why? How did you train and evaluate them?

(d) **Results and Discussion:** What did your model achieve? What do the results mean in the context of the problem?

(e) **Conclusion and Recommendations:** What is the main insight? What would you suggest the business or organisation do next?

(f) **References:** Cite all datasets, libraries, papers, and tools used. Use APA 7th Edition. Use in-text citations wherever required.